

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12406848

Kind Code

B2

Date of Patent

September 02, 2025

Inventor(s)

Wu; Ming-Yuan et al.

Method of fabricating a gate cut feature for multi-gate semiconductor devices

Abstract

A method includes forming a fin protruding from a substrate, forming a first dielectric feature adjacent to the fin over the substrate, forming a cladding layer over the fin and the first dielectric feature, and removing a portion of the cladding layer to form an opening. The opening exposes the first dielectric feature. The method further includes forming a second dielectric feature adjacent to the cladding layer, the second dielectric feature filling the opening, forming a dummy gate stack over the fin and the second dielectric feature, forming source/drain (S/D) features in the fin adjacent to the dummy gate stack, and replacing the dummy gate stack and the cladding layer with a metal gate stack. The second dielectric feature divides the metal gate stack.

Inventors: Wu; Ming-Yuan (Hsinchu, TW), Lin; Da-Wen (Hsinchu, TW), Fu; Yi-Ting (Hsinchu, TW), Cheng; Hsu-Chieh (Hsinchu, TW), Jiao; Min (Hsinchu, TW)

Applicant: Taiwan Semiconductor Manufacturing Company, Ltd. (Hsin-Chu, TW)

Family ID: 84313957

Assignee: TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD (Hsinchu, TW)

Appl. No.: 17/736898

Filed: May 04, 2022

Prior Publication Data

Document Identifier

US 20230015372 A1

Publication Date

Jan. 19, 2023

Related U.S. Application Data

us-provisional-application US 63222538 20210716

Publication Classification

Int. Cl.: H01L21/28 (20250101); H01L21/02 (20060101); H10D30/01 (20250101); H10D30/67 (20250101); H10D62/10 (20250101); H10D64/01 (20250101); H10D84/01 (20250101); H10D84/03 (20250101); H10D84/83 (20250101)

U.S. Cl.:

CPC H01L21/28123 (20130101); H01L21/0259 (20130101); H10D30/031 (20250101); H10D30/6735 (20250101); H10D30/6757 (20250101); H10D62/118 (20250101); H10D64/017 (20250101); H10D84/0128 (20250101); H10D84/0151 (20250101); H10D84/038 (20250101); H10D84/83 (20250101);

Field of Classification Search

CPC: H01L (29/42392); H01L (29/6645); H01L (29/66439); H01L (29/78696); H01L (29/0665); H01L (29/66742); H01L (21/823481); H01L (21/28123); H01L (21/0259); H01L (21/823412); H01L (27/088)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8772109	12/2013	Colinge	N/A	N/A
8785285	12/2013	Tsai et al.	N/A	N/A
8816444	12/2013	Wann et al.	N/A	N/A
8823065	12/2013	Wang et al.	N/A	N/A
8860148	12/2013	Hu et al.	N/A	N/A
9105490	12/2014	Wang et al.	N/A	N/A
9236267	12/2015	De et al.	N/A	N/A
9236300	12/2015	Liaw	N/A	N/A
9520482	12/2015	Chang et al.	N/A	N/A
9576814	12/2016	Wu et al.	N/A	N/A
2020/0006075	12/2019	Wang et al.	N/A	N/A
2021/0234017	12/2020	Yao	N/A	H01L 21/02603
2022/0310783	12/2021	Chen	N/A	H01L 29/0653
2022/0406900	12/2021	Wu	N/A	H01L 29/66439

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
202117855	12/2020	TW	N/A

Primary Examiner: Chambliss; Alonzo

Attorney, Agent or Firm: HAYNES AND BOONE, LLP

Background/Summary

PRIORITY DATA (1) This application claims priority to U.S. Provisional Patent Application No. 63/222,538 filed on Jul. 16, 2021, the entire disclosure of which is incorporated herein by reference.

BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs and, for these advancements to be realized, similar developments in IC processing and manufacturing are needed.

(2) As integrated circuit (IC) technologies progress towards smaller technology nodes, the size of the gate cut feature is reduced accordingly to achieve the design dimensions. While methods of fabricating the gate cut feature have been generally adequate for their intended purposes, they have not been entirely satisfactory in all aspects. For example, multiple layers of different materials need to be removed when forming a trench for the gate cut feature, which requires multiple etching processes. In addition, the low aspect ratio of the trench in smaller technology nodes lead to difficulties during the filling of the gate cut feature in the trench, such as incomplete filling caused by trapped air bubbles. It is a need to improve the gate cut feature and the method of fabricating the same.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates a flowchart of an example method for fabricating a semiconductor device according to various embodiments of the present disclosure.

(3) FIG. 2A is a three-dimensional perspective view of a portion of an example semiconductor device according to various embodiments of the present disclosure.

(4) FIG. 2B is a planar top view of the semiconductor device shown in FIG. 2A according to various embodiments of the present disclosure.

(5) FIGS. 3A, 4A, 5A, 5A-1, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, and 14A are planar top views of the semiconductor device as shown in FIGS. 2A and/or 2B during intermediate stages of the method shown in FIG. 1 according to various embodiments of the present disclosure.

(6) FIGS. 3B, 4B, 5B, 5B-1, 6B, 6B-1, 7B, 8B, 9B, 10B, 11B, 12B, 13B, and 14B are cross-sectional views of the semiconductor device taken along line BB' as shown in FIGS. 2A and/or 2B during intermediate stages of the method shown in FIG. 1 according to various embodiments of the present disclosure.

(7) FIGS. 9C, 10C, 11C, 12C, 13C, 14C, and 14C-1 are cross-sectional views of the semiconductor device taken along line CC' as shown in FIGS. 2A and/or 2B during intermediate stages of the method shown in FIG. 1 according to various embodiments of the present disclosure.

DETAILED DESCRIPTION

(8) The following disclosure provides many different embodiments, or examples, for implementing different features of the disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a feature on, connected to, and/or coupled to another feature in the present disclosure that follows may include embodiments in which the features are formed in direct contact, and may also include embodiments in which additional features may be formed interposing the features, such that the features may not be in direct contact. In addition, spatially relative terms, for example, “lower,” “upper,” “horizontal,” “vertical,” “above,” “over,” “below,” “beneath,” “up,” “down,” “top,” “bottom,” etc. as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) are used for ease of the present disclosure of one features relationship to another feature. The spatially relative terms are intended to cover different orientations of the device including the features.

(9) Furthermore, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range including the number described, such as within $\pm 10\%$ of the number described or other values as understood by person skilled in the art. For example, the term “about 5 nanometers” encompasses the dimension range from 4.5 nanometers to 5.5 nanometers. Still further, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(10) The present disclosure is generally related to semiconductor devices, and more particularly to field-effect transistors (FETs), such as three-dimensional nanostructure (NS) FETs (alternatively referred to as gate-all-around, or GAA FETs), in memory and/or standard logic cells of an integrated circuit (IC) structure. Generally, an NS FET includes a plurality of active regions formed by vertically stacked nanosheets, source/drain (S/D) features disposed in the NS FET, and metal gate stacks disposed adjacent to the S/D features and interleaving with the nanosheets. In such configuration, the metal gate stacks are truncated (or cut, separated) by gate cut features to control various active regions according to design requirements. However, in some instances, the forming of the gate cut feature are complex and troublesome when the size of the IC structure becomes smaller. For example, the forming of the trench for gate cut feature requires multiple etching processes to remove different layers of materials. In addition, the low aspect ratio (a width along Y direction: a length along Z direction) of the trench may cause air bubbles and incomplete filling of the gate cut feature in the trench. The present embodiments solve the aforementioned issues. The present disclosure includes multiple embodiments. Different embodiments may have different advantages, and no particular advantage is necessarily required of any embodiment.

(11) Referring now to FIG. 1, a flowchart of a method **100** of forming a semiconductor device **200** (hereafter simply referred to as the device **200**) is illustrated according to various aspects of the present disclosure. The method **100** is merely an example and not intended to limit the present disclosure beyond what is explicitly recited in the claims. Additional operations can be provided before, during, and after the method **100**, and some operations described can be replaced, eliminated, or moved around for additional embodiments of the method. The method **100** is described below in conjunction with FIGS. 3A to 14C-1, which are various top planar views and cross-sectional of the device **200** as shown in FIGS. 2A and 2B at intermediate steps of the method **100**. For examples, 3A, 4A, 5A, 5A-1, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, and 14A are planar top views of the device **200** as shown in FIGS. 2A and/or 2B; FIGS. 3B, 4B, 5B, 5B-1, 6B, 6B-1, 7B, 8B, 9B, 10B, 11B, 12B, 13B, and 14B are cross-sectional views of the device **200** taken along line BB' as shown in FIGS. 2A and/or 2B; and FIGS. 9C, 10C, 11C, 12C, 13C, 14C, and 14C-1 are cross-sectional views of the device **200** taken along line CC' as shown in FIGS. 2A and/or 2B.

(12) The device **200** may be an intermediate device fabricated during processing of an IC, or a portion thereof, that may comprise static random-access memory (SRAM) and/or other logic

circuits, passive components such as resistors, capacitors, and inductors, and active components such as NS FETs, FinFETs, metal-oxide semiconductor field effect transistors (MOSFETs), complementary metal-oxide semiconductor (CMOS) transistors, bipolar transistors, high voltage transistors, high frequency transistors, and/or other transistors. In the present embodiments, the device **200** includes one or more NS FETs. The present disclosure is not limited to any particular number of devices or device regions, or to any particular device configurations. Additional features can be added to the device **200**, and some of the features described below can be replaced, modified, or eliminated in other embodiments of the device **200**.

(13) At operations **102** and **104**, referring to FIGS. **2A**, **2B**, **3A**, and **3B**, the method **100** forms the device **200** that includes multiple three-dimensional fin active regions (hereafter referred to as fins) **204** protruding from a semiconductor substrate (hereafter referred to as the substrate) **202**, where the fins **204** are separated by isolation features (or dielectric feature) **208**. Each portions of the isolation features **208** is defined by a width $w_{sub.1}$ that equals to a distance between two adjacent fins **204**, where $w_{sub.1}$ is about 30 nm to about 50 nm. Portions of the fins **204** are denoted to as **204A**, **204B**, and **204C**, portions of the isolation features **208** are denoted as **208A** (between fins **204A** and **204B**) and **208B** (between fins **204B** and **204C**) for clarity and simplicity in the present disclosure. The denotations of the fins **204** and the isolation features **208**, by themselves, do not indicate any differences therebetween.

(14) The substrate **202** may include an elemental (single element) semiconductor, such as silicon (Si), germanium (Ge), and/or other suitable materials; a compound semiconductor, such as silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, indium antimonide, and/or other suitable materials; an alloy semiconductor, such as SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, and/or other suitable materials. The substrate **202** may be a single-layer material having a uniform composition. Alternatively, the substrate **202** may include multiple material layers having similar or different compositions suitable for IC device manufacturing.

(15) In the present embodiments, still referring to FIGS. **2A**, and **3B**, each fin **204** (including the fins **204A**, **204B**, and **204C**) includes a multi-layer structure (ML) of alternating non-channel layers (or sacrificial layers) **205** and channel layers **206** stacked vertically over the substrate **202**. A hard mask **207** may be formed over the ML. In some embodiments, the hard mask layer **207** has a composition the same as or similar to the non-channel layer **205**. In some embodiments, an oxide layer **210** may be optionally formed between the hard mask **207** and the topmost channel layer **206**.

(16) In the present embodiments, the non-channel layers **205** are sacrificial layers configured to be removed at a subsequent processing step, thereby providing openings between the channel layers **206** for forming the metal gate stack therein. Each channel layer **206** may include a semiconductor material such as, for example, Si, Ge, SiC, SiGe, GeSn, SiGeSn, SiGeCSn, other suitable semiconductor materials, or combinations thereof, while each non-channel layer **205** has a composition different from that of the channel layer **206**. In one such example, the channel layer **206** may include elemental Si and the non-channel layer **205** may include SiGe. In another example, the channel layers **206** may include elemental Si and the non-channel layers **205** may include elemental Ge. In the present embodiments, the channel layers **206** include elemental Si and the non-channel layers **205** include SiGe. In some examples, each fin **204** may include a total of three to ten pairs of alternating non-channel layers **205** and channel layers **206**; of course, other configurations may also be applicable depending upon specific design requirements.

(17) In the present embodiments, forming the ML includes alternately growing the non-channel layers **205** and the channel layers **206** in a series of epitaxy processes. The epitaxy processes may be implemented by chemical vapor deposition (CVD) techniques (for example, vapor-phase epitaxy (VPE), ultra-high vacuum CVD (UHV-CVD), low-pressure (LP-CVD), and/or plasma-enhanced CVD (PE-CVD)), molecular beam epitaxy, other suitable selective epitaxial growth (SEG) processes, or combinations thereof. The epitaxy process may use gaseous and/or liquid

precursors containing a suitable material (e.g., Ge for the non-channel layers **205**), which interact with the composition of the underlying substrate, e.g., the substrate **202**. In some examples, the non-channel layers **205** and the channel layers **206** may be formed into nanosheets, nanowires, or nanorods. A sheet (or wire) release process may then be implemented to remove the non-channel layers **205** to form openings between the channel layers **206**, and a metal gate stack is subsequently formed in the openings, thereby providing an NS FET.

(18) In the present embodiments, the fins **204** are fabricated from the ML using a series of photolithography and etching processes. For example, the photolithography process may include forming a photoresist layer overlying the ML, exposing the photoresist layer to a pattern, performing post-exposure bake processes, and developing the exposed photoresist layer to form a patterned masking element (not depicted). The ML is then etched using the patterned masking element as an etch mask, thereby leaving the fins **204A**, **204B**, and **204C** protruding the substrate **202** and separated by trenches **203**. The etching process may include dry etching, wet etching, reactive ion etching (RIE), other suitable processes, or combinations thereof. The patterned masking element is subsequently removed from the ML using any suitable process, such as ashing and/or resist stripping.

(19) In some embodiments, the hard mask **207** is configured to protect the fins **204** (including the fins **204A**, **204B**, and **204C**) during subsequent processing steps and is later removed from the device **200**. In some embodiments, the hard mask **207** has a composition similar to or the same as that of the non-channel layers **205** and includes, for example, SiGe. In some embodiments, the optional oxide layer **210** includes any suitable materials, such as silicon oxide (SiO_{2-x} , $1 \leq x \leq 2$). In some embodiments, the hard mask **207** and the oxide layer **210** are collectively patterned with the ML during the fin fabrication process.

(20) The isolation features **208** (including the isolation features **208A** and **208B**) may include silicon oxide (SiO_{2-x} , $1 \leq x \leq 2$), tetraethylorthosilicate (TEOS), doped silicon oxide (e.g., borophosphosilicate glass (BPSG), fluoride-doped silicate glass (FSG), phosphosilicate glass (PSG), boron-doped silicate glass (BSG), etc.), a low-k dielectric material (having a dielectric constant less than that of silicon oxide, which is about 3.9), other suitable materials, or combinations thereof. In one embodiment, the isolation features **208** are formed by filling trenches **203** between the fins **204** with a dielectric material described above by any suitable method, such as CVD, flowable CVD (FCVD), spin-on-glass (SOG), other suitable methods, or combinations thereof. The dielectric material may subsequently be planarized by a chemical-mechanical planarization/polishing (CMP) process and selectively etched back to form the isolation features **208**. In some embodiments, the isolation features **208** include shallow trench isolation (STI) features. In some embodiments, the isolation features **208** include a single-layer structure or a multi-layer structure. As depicted in FIG. 3B, top surfaces of the isolation features **208** are exposed in the trenches **203**.

(21) At operation **106**, referring to FIGS. 4A and 4B, the method **100** forms a cladding layer **209** along sidewalls of the fins **204** and over the isolation features **208**. In some embodiments, the cladding layer **209** has a composition that is similar to that of the non-channel layers **205**, such that the cladding layer **209** and the non-channel layers **205** can be removed together. In one example, the cladding layer **209** and the non-channel layers **205** both include silicon germanium (SiGe) but different in germanium concentrations, such as the cladding layer **209** may have a higher molar ratio of germanium. In some embodiments, the cladding layer **209** is deposited epitaxially by a suitable method discussed above with respect to forming the ML. In alternative embodiments, the cladding layer **209** is deposited conformally, rather than grown epitaxially, over surfaces of the device **200** as an amorphous layer, such that the cladding layer **209** is also formed over the isolation features **208**. In the present embodiments, the cladding layer **209** is formed to a thickness t_i ranges from about 5 nm to about 10 nm. The cladding layer **209** partially fills the trenches **203**. Each trench **203** has a width $w_{sub.2}$ between two adjacent fins **204** after the deposition of the cladding

layer **209**, where the width $w_{sub.2}$ is about 10 nm to about 40 nm.

(22) Generally, the gate cut feature is formed after the forming of the dummy gate stacks or after the forming of the metal gate stacks. In these cases, trenches are formed by removing portions of the dummy gate stack or the metal gate stacks, the layers (e.g., the interlayer dielectric (ILD) layer disposed thereover, the isolation features between the fins, and/or the substrate. The gate cut features are then formed in the trenches by filling proper materials. However, the forming of the trenches requires to remove multilayers of different materials, which may involve multiple complex etching processes. In addition, the aspect ratios of the trenches become higher as the scaling down of the IC device. The low aspect ratio may cause issues during the forming of the gate cut feature therein. For example, air bubbles may be trapped in the trench and lead to incomplete filling of the trench. The embodiments of the present disclosure solve these issues. For example, the present embodiments forms gate cut features with a low aspect ratio before the forming of the dummy gate stacks, thereby avoiding multiple etching processes. The gate cut features are disposed between fins and contact the isolation features through an opening in the cladding layer of the fins. Detail of the present embodiments is explained below with respect to FIG. 1 and FIG. 5A to FIG. 14C-1.

(23) At operation **108**, referring to FIGS. 5A and 5B, the method **100** performs an etching process **502** to selectively remove portions of the cladding layer **209** between the fins **204A** and **204B**, thereby forming an opening **215** and exposing portions of the isolation features **208A**. The isolation features **208A**, the vertical portions of the cladding layer **209**, and the horizontal portion of the cladding layer **209** disposed over the fins **204** (or the hard mask **207** if present) remain intact or substantially intact. The isolation feature **208B** remains covered under the cladding layer **209**. The opening **215** separates (or truncates, divides, cuts) the cladding layer **209** into a first portion over the fin **204A** and a second portion over the fins **204B** and **204C**. In some embodiments, the etching process **502** includes a dry etching process, an RIE process, or combinations thereof. In some embodiments, a photolithography may be used to remove the portions of the cladding layer **209**.

(24) As depicted in FIGS. 5A and 5B, the etching process removes a portion of the horizontal portion of the cladding layer **209** defined by a width $w_{sub.3}$, where the width $w_{sub.3}$ is about 1 to about 2 times of the thickness t_i to ensure the aspect ratio of the opening **215** is suitable for filling dielectric materials therein as explained in detail below. Each of the trenches **203** is defined by the width $w_{sub.2}$ at a top portion and a width $w_{sub.3}$ at the bottom portion. In some embodiments, the width $w_{sub.3}$ is greater than 80% of the width $w_{sub.2}$ but less than $w_{sub.2}$. If the width $w_{sub.3}$ is less than $xx\%$, the gate cut feature formed in subsequent processes may not land stably on the isolation feature **208A**. In one example, the width $w_{sub.3}$ ranges from about 20 nm to about 30 nm. Remaining portions of the cladding layer **209** forms a L shape or a reverse L shape over the isolation feature **208A** and the sidewalls of the fins **204A** and **204B**, respectively.

(25) FIGS. 5A-1 and 5B-1 depict alternative embodiments of these depicted in FIGS. 5A and 5B, where the etching process **502** removes the entire horizontal portion of the cladding layer **209** over the isolation feature **208A**, thereby enabling further reduction in sizes of the device **200**. In this case, the width $w_{sub.3}$ equals to the width $w_{sub.2}$. The trench **203** has a uniform width $w_{sub.2}$ throughout the entire length of the trench **203** along Z direction.

(26) At operation **110**, referring to FIGS. 6A and 6B, the method **100** forms a gate cut feature (or dielectric feature) **211** in one of the trenches **203** between the fins **204A** and **204B**, thereby completely filling the opening **215** and the trench **203**. The gate cut feature **211** is configured to truncate (or isolate, separate, cut) the metal gate stack formed in subsequent processes explained in detail below. A bottom surface of the gate cut feature **211** contacts a top surface of the isolation feature **208A**.

(27) In the embodiment depicted in FIG. 6B, the gate cut feature **211** includes a top portion defined by the width $w_{sub.2}$ and a bottom portion defined by the width $w_{sub.3}$. The bottom portion is formed in the opening **215** and cuts the cladding layer **209** (and thereby the later formed metal gate

stack). FIG. 6B-1 depicts an alternative embodiment of that depicted in FIG. 6B, where the gate cut feature **211** is defined by the width $w_{\text{sub.2}}$ throughout the entire length along Z direction. FIG. 6B corresponds to the embodiment depicted in FIG. 5B, and FIG. 6B-1 corresponds to the embodiment depicted in FIG. 5B-1. Both the embodiments depicted in FIGS. 6B and 6B-1 undergo the same process hereafter, therefore the embodiments depicted in FIG. 6B is used as an example to illustrate the operations of the method **100** and the related embodiments.

(28) The gate cut feature **211** may include any suitable material, such as silicon oxide ($\text{SiO}_{\text{sub.x}}$, $1 \leq x \leq 2$), silicon nitride (SiN), silicon carbide (SiC), oxygen-containing silicon nitride (SiON), oxygen-containing silicon carbide (SiOC), carbon-containing silicon nitride (SiCN), aluminum oxide ($\text{Al}_{\text{sub.2O.sub.3}}$), tetraethylorthosilicate (TEOS), doped silicon oxide (e.g., borophosphosilicate glass (BPSG), FSG, phosphosilicate glass (PSG), boron-doped silicate glass (BSG), etc.), a high-k dielectric material (having a k value greater than that of silicon oxide, which is about 3.9), other suitable materials, or combinations thereof. The high-k dielectric material may include oxygen, lanthanum, aluminum, titanium, zirconium, tantalum, other suitable materials, or combinations thereof. For example, the high-k dielectric material may include hafnium oxide ($\text{HfO}_{\text{sub.2}}$), lanthanum oxide ($\text{La}_{\text{sub.2O.sub.3}}$), other high-k oxide materials, or combinations thereof. In the present embodiments, the gate cut feature **211** includes a material that is the same as a material included in the isolation feature **208**. In one such embodiment, the gate cut feature **211** includes silicon oxide.

(29) The gate cut feature **211** may be single-layered or multi-layered. In some embodiments, the gate cut feature **211** each include a lining layer **213** and a filling layer **201** different from the lining layer **213** in composition. In some embodiments, the lining layer **213** has a higher density than that of the filling layer **201**. In some embodiments, the filling layer **201** has a lower dielectric constant than the lining layer **213**. In some embodiments, the lining layer **213** is formed conformally along sidewalls of the cladding layer **209** and over the isolation features **208A** prior to the deposition of the filling layer **201**. The lining layer **213** may be formed by any suitable method, such as ALD, CVD, other suitable methods, or combinations thereof. The filling layer **201** is subsequently formed over the lining layer **213** by any suitable method, such as CVD, FCVD, other suitable methods, or combinations thereof. In some embodiments, the device **200** is subsequently planarized by one or more CMP process to expose a top surface of the hard mask **207**.

(30) The dummy gate cut feature (or dielectric feature) **211'** is formed as a byproduct in the same process that forms the gate cut feature **211**. The dummy gate cut feature **211'** does not truncate, or completely truncate, the metal gate stack formed in subsequent processes. However, additional patterning processes are required to avoid the forming of the dummy gate cut feature **211'** during the forming of the gate cut feature **211**. In the present embodiments, a top surface of the dummy gate cut feature **211'** is coplanar with a top surface of the gate cut feature **211**, while a bottom surface of the dummy gate cut feature **211'** is above a bottom surface gate cut feature **211**. The dummy gate cut feature **211'** is separated from the isolation feature **208B** by the cladding layer **209**. The cladding layer **209** is continuous between the dummy gate cut feature **211'** and the isolation feature **208B**. The dummy gate cut feature **211'** is defined by the width $w_{\text{sub.2}}$ throughout the entire length along Z direction.

(31) The dummy gate cut feature **211'** includes the same materials as these included in the gate cut feature **211**. The same as the gate cut feature **211**, the dummy gate cut feature **211'** may include one or more layers. In the present embodiments, the dummy gate cut feature **211'** includes a lining layer **201'** and a filling layer **213'**. The lining layer **201'** and the filling layer **213'** are formed in the same processes that the lining layer **201** and the filling layer **213** are formed in. The filling layer **213'** has the same size as the filling layer **213**. A top surface and a bottom surface of the filling layer **213'** are coplanar with a top surface and a bottom surface of the filling layer **213**, respectively.

(32) At operation **112**, referring to FIGS. 7A, 7B, 8A and 8B, the method **100** forms the dielectric helmets **214** over the gate cut feature **211** and the dummy gate cut feature **211'**. The dielectric

helmet over the gate cut feature **211** is denoted as **214A** and the dielectric helmet over the dummy gate cut feature **211'** is denoted as **214B** for easier reference and does not indicate the differences therebetween except what explicitly stated in the disclosure. As depicted in FIGS. 7A and 7B, the method **100** first recesses top portions of the gate cut feature **211** and the dummy gate cut feature **211'** in an etching process **702** to form trenches **212**. The etching process **702** may include any suitable process, such as a dry etching process, a wet etching process, an RIE process, other suitable processes, or combinations thereof.

(33) Referring to FIGS. 8A and 8B, the method **100** fills the trenches **212** with the dielectric helmet **214** (including **214A** and **214B**) in process **802**. The dielectric helmets **214** are oriented lengthwise parallel to the lengthwise direction of the fins **204** and are separated from the sidewalls of the fins **204** by the cladding layer **209**. In the present embodiments, the dielectric helmets **214** have an etching selectivity with respect to the gate cut feature **211**, the dummy gate cut feature **211'**, and the cladding layer **209**. In the present embodiments, at least a portion of the dielectric helmets **214** (e.g., the dielectric helmet **214A**) is configured to provide isolation for a subsequently-formed metal gate stack over the fins **204**. In other words, at least a portion of the dielectric helmets **214** is configured to truncate (or cut, separate) the metal gate stack into multiple portions.

(34) In some embodiments, the dielectric helmets **214** include silicon nitride (SiN), silicon carbide (SiC), oxygen-containing silicon nitride (SiON), oxygen-containing silicon carbide (SiOC), carbon-containing silicon nitride (SiCN), aluminum oxide (Al.sub.2O.sub.3), tetraethylorthosilicate (TEOS), doped silicon oxide (e.g., borophosphosilicate glass (BPSG), FSG, phosphosilicate glass (PSG), boron-doped silicate glass (BSG), etc.), a high-k dielectric material (having a k value greater than that of silicon oxide, which is about 3.9), other suitable materials, or combinations thereof. The high-k dielectric material may include oxygen, lanthanum, aluminum, titanium, zirconium, tantalum, other suitable materials, or combinations thereof. For example, the high-k dielectric material may include hafnium oxide (HfO.sub.2), lanthanum oxide (La.sub.2O.sub.3), other high-k oxide materials, or combinations thereof. The dielectric helmets **214** can be single layered or multi-layered. In the present embodiments, the dielectric helmets **214** are each a single layer structure with uniform composition.

(35) At operation **114**, referring to FIGS. 9A, 9B, and 9C, the method **100** forms one or more dummy gate stacks **220** over channel regions of the fins **204**, the dielectric helmets **214**, and the cladding layer **209**. Each dummy gate stack **220** may include a dummy gate electrode (not depicted separately) disposed over an optional dummy gate dielectric layer and/or an interfacial layer. In the present embodiments, the dummy gate stacks **220** are later replaced with metal gate stacks, which are truncated (or cut) by the gate cut feature **211** and/or the dielectric helmets **214**. The dummy gate stacks **220** may be formed by a series of deposition and patterning processes. For example, the dummy gate stacks **220** may be formed by depositing a polysilicon (poly-Si) layer over the substrate **202**, and subsequently patterning the poly-Si layer via a series of photolithography and etching processes. To accommodate the patterning process and protect the dummy gate stacks **220** during subsequent fabrication processes, a hard mask layer (not depicted) may be formed over the dummy gate stacks **220**.

(36) The method **100** may form top gate spacers (not shown) over sidewalls of the dummy gate stack **220**. The top gate spacers may be a single-layer structure or a multi-layer structure and may include silicon oxide, SiN, SiC, SiON, SiOC, SiCN, air, a low-k dielectric material, a high-k dielectric material (e.g., hafnium oxide (HfO.sub.2), lanthanum oxide (La.sub.2O.sub.3), etc.), other suitable materials, or combinations thereof. The top gate spacers may be formed by first depositing a dielectric layer over the dummy gate stacks **220** via a suitable deposition method (e.g., CVD and/or ALD) and subsequently removing portions of the dielectric layer in an anisotropic (e.g., directional) etching process (e.g., a dry etching process), leaving the top gate spacers on the sidewalls of the dummy gate stacks **220**.

(37) The method **100** then removes portions of the hard mask **207** and the oxide layer **210** in S/D

regions of the fins **204** to expose the topmost channel layer **206**, while the portions of the hard mask **207** and the oxide layer **210** under the dummy gate stacks **220** remain intact. In some embodiments, the removing of the hard mask **207** utilizes an etching process including a dry etching process, an RIE process, or combinations thereof.

(38) The method **100** subsequently forms S/D recesses (not shown) in the fins **204** adjacent to the dummy gate stacks **220**. In the present embodiments, the method **100** implements an etching process that selectively removes portions of the fins **204** in the S/D regions without removing, or substantially removing, the dummy gate stacks **220**, the dielectric helmets **214**, and the cladding layer **209**. In some embodiments, the etching process is a dry or wet etching process employing a suitable etchant capable of removing Si (i.e., the channel layers **206**) and SiGe (i.e., the non-channel layers **205**) of the ML. In some non-limiting examples, a dry etchant may be a chlorine-containing etchant including Cl.sub.2, SiCl.sub.4, BCl.sub.3, other chlorine-containing gas, or combinations thereof. A cleaning process may subsequently be performed to clean the S/D recesses with a hydrofluoric acid (HF) solution or other suitable solution.

(39) The method **100** then recesses the cladding layer **209** in the S/D recesses, such that top surfaces of the cladding layer is below the top surfaces of the gate cut feature **211**. Remaining portions of the cladding layer in the S/D recesses serve as spacers for forming the S/D features in subsequent processes. In the present embodiments, the remaining portions of the cladding layer **209** may be L shaped and are disposed along sidewalls and bottom surfaces of the gate cut feature **211**. Alternatively, the remaining portions of the cladding layer **209** may be U shaped and wrapping around a bottom portion of the dummy gate cut feature **211'**. In some embodiments, the portions of the cladding layer **209** is removed by an etching process including a dry etching process, an RIE process, or combinations thereof.

(40) Thereafter, the method **100** forms inner gate spacers (not shown) on sidewalls of the non-channel layers **205** that are exposed in the S/D recesses. The inner gate spacers may be a single-layer structure or a multi-layer structure and may include silicon oxide, SiN, SiCN, SiOC, SiON, SiOCN, a low-k dielectric material, air, a high-k dielectric material, hafnium oxide (HfO.sub.2), lanthanum oxide (La.sub.2O.sub.3), other suitable dielectric material, or combination thereof. In some embodiments, the inner gate spacers have a composition different from that of the top gate spacers. Forming the inner gate spacers includes performing a series of etching and deposition processes. For example, forming the inner gate spacers may begin with selectively removing portions of the non-channel layers **205** without removing, or substantially removing, portions of the channel layers **206** to form trenches (not depicted). The non-channel layers **205** may be etched by a dry etching process. Subsequently, one or more dielectric layers are deposited in the trenches by any suitable method, such as ALD, CVD, PVD, other suitable methods, or combinations thereof, followed by one or more etching processes to remove (i.e., etch back) excess dielectric layer(s) deposited on surfaces of the channel layers **206** that are exposed in the S/D recesses, thereby forming the inner gate spacers.

(41) At operation **116**, referring to FIGS. **10A**, **10B**, and **10C**, the method **100** forms S/D features **224** in each of the S/D recesses. The S/D features **224** may be suitable for forming a p-type FET device (i.e., including a p-type epitaxial material) or, alternatively, an n-type FET device (i.e., including an n-type epitaxial material). The p-type epitaxial material may include one or more epitaxial layers of silicon germanium (epi SiGe) each doped with a p-type dopant such as boron, germanium, indium, gallium, other p-type dopants, or combinations thereof. The n-type epitaxial material may include one or more epitaxial layers of silicon (epi Si) or silicon carbon (epi SiC) each doped with an n-type dopant such as arsenic, phosphorus, other n-type dopants, or combinations thereof. In some embodiments, one or more epitaxy growth processes are performed to grow an epitaxial material in each S/D recess and over the inner gate spacers. For example, the method **100** may implement an epitaxy growth process similar to that discussed above with respect to forming the ML. In some embodiments, the epitaxial material is doped in-situ by adding a

dopant to a source material during the epitaxial growth process. In some embodiments, the epitaxial material is doped by an ion implantation process after performing the deposition process. In some embodiments, an annealing process is subsequently performed to activate the dopants in the S/D features **224**.

(42) Subsequently, still referring to FIGS. **10A**, **10B**, and **10C**, the method **100** forms an ILD layer **242** over the device **200**, thereby filling the space between adjacent dummy gate stacks **220**. The ILD layer **242** may include silicon oxide, a low-k dielectric material, TEOS, doped silicon oxide (e.g., BPSG, FSG, PSG, BSG, etc.), other suitable dielectric materials, or combinations thereof, and may be formed by any suitable method, such as CVD, FCVD, SOG, other suitable methods, or combinations thereof. The method **100** subsequently performs one or more CMP process to expose top surfaces of the dummy gate stacks **220**.

(43) The method **100** may also form a contact etch-stop layer (CESL) (not shown) over the device **200** prior to the forming of the ILD layer **242** to protect the underlying components, such as the S/D features **224**, during subsequent fabrication processes. The CESL may include any suitable dielectric material, such as SiN, SiCN, other suitable materials, or combinations thereof, and may be formed by CVD, ALD, physical vapor deposition (PVD), other suitable methods, or combinations thereof. In the present embodiments, the CESL provides etching selectivity with respect to its surrounding dielectric components to ensure protection against inadvertent damage to these components.

(44) At operation **118**, referring to FIGS. **11A**, **11B**, and **11C**, the method **100** removes portions of the dielectric helmets **214** in a patterning process. Remaining portions of the dielectric helmets serve as extensions of the gate cut feature **211** to truncate (or separate, cut) top portions of the metal gate stack. The patterning of the dielectric helmets **214** includes a photolithography process to remove a portion of the dummy gate stack **220** to form an opening **243**, thereby exposing the dielectric helmet **214B** to be removed. It is noted that the dielectric helmet **214B** may have a dimension smaller than the line width limit of the photolithography process. In this regard, the opening **243** is wider than the dielectric helmet **214B** to accommodate the dimension of the dielectric helmet **214B**. The dielectric helmets **214** may be removed in a wet etching, a dry etching, or a combination thereof. Remaining portions of the dummy gate stacks **220** are used as a mask in the etching processes.

(45) At operation **120**, referring to FIGS. **12A** to **13C**, the method **100** replaces the dummy gate stacks **220**, the cladding layer **209** under the dummy gate stack **220**, and the non-channel layers **205** with metal gate stacks **230**, where the metal gate stacks **230** are separated by the gate cut feature **211** and the dielectric helmet **214A**. Prior to the operation **120**, the remaining hard mask **207** and the oxide layer **210** are removed in one or more etching processes. The method **100** then removes the remaining portions of the dummy gate stacks **220** to form gate trenches; removes the cladding layer **209** exposed therein; and removes the non-channel layers **205** from the ML to form openings between the channel layers **206**, as depicted in FIG. **12C**. The method **100** subsequently forms metal gate stacks **230** in the gate trenches and the openings, such that each metal gate stack **230** wraps around (or is interleaved with) each channel layer **206** as depicted in FIGS. **13A** to **13C**.

(46) In the present embodiments, the metal gate stack **230** includes a gate dielectric layer (not depicted separately) and a metal gate electrode (not depicted separately) over the gate dielectric layer. The gate dielectric layer may include a high-k dielectric material, such as HfO₂, La₂O₃, other suitable materials, or combinations thereof. The metal gate electrode includes at least one work function metal layer and a bulk conductive layer disposed thereover. The work function metal layer may be a p-type or an n-type work function metal layer. Example work function metals include TiN, TaN, WN, ZrSi₂, MoSi₂, TaSi₂, NiSi₂, Ti, TaAl, TaAlC, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, other suitable work function metals, or combinations thereof. The bulk conductive layer may include Cu, W, Al, Co, Ru, other suitable materials, or combinations thereof. The metal gate stack **230** may further include other material layers (not

depicted), such as an interfacial layer disposed on surfaces of the channel layers **206**, a capping layer, a barrier layer, other suitable layers, or combinations thereof. Various layers of the metal gate stack **230** may be formed by various methods, including ALD, CVD, PVD, plating, other suitable methods, or combinations thereof. After forming the bulk conductive layer, one or more CMP processes are performed to remove excessive material formed on top surface of the ILD layer **242**, thereby planarizing the device **200**.

(47) Thereafter, referring to FIG. **14A** to **14C**, the method **100** recesses the metal gate stack **230**, such that a top surface of the recessed metal gate stack **230** is below a top surface of the dielectric helmet **214A**. The dielectric helmets **214A**, together with the gate cut feature **211**, truncates (or separate) the recessed metal gate stacks **230** into metal gate stack **230A** and **230B** as depicted in FIG. **14C**. In other words, the metal gate stack **230A** is separated from the metal gate stack **230B** by the gate cut feature **211** at the bottom and by the dielectric helmet **214** at the top. In some embodiments, the etching process includes any suitable method, such as ALD, CVD, PVD, other suitable methods, or combinations thereof.

(48) The metal gate stack **230B** wraps around the dummy gate cut feature **211'**, where a portion of the metal gate stack **230B** spans from the fin **204B** to the fin **204C** over the dummy gate cut feature **211'**. The metal gate stack **230B** is continuous under the dummy gate cut feature **211'**. The dummy gate cut feature **211'** is separated from the isolation feature **208B** by the metal gate stack **230B**. In other words, the dummy gate cut feature **211'** is embedded in the metal gate stack **230B** and does not truncate (or cut, separate) the metal gate stack **230B**. The portions of the metal gate stack **230B** over and under the dummy gate cut feature ensures the connection and reduces resistance of the metal gate stack **230B**.

(49) In alternative embodiments, referring to FIGS. **1** and **14C-1**, the operations **112** and **118** may be omitted from the method **100** to simplify the fabricating processes. In this case, the forming of the dielectric helmets **214** is omitted. Accordingly, instead of recessing the top surfaces of the metal gate stacks **230** below the dielectric helmets **214** and above the top surface of the gate cut feature **211**, the method **100** recesses the top surface of the metal gate stack **230** to be coplanar with the top surface of the gate cut feature **211**. As such, the gate cut feature **211** truncates the metal gate stack **230** into metal gate stacks **230A** and **230B** without the help of the dielectric helmets **214** as depicted in FIG. **14C**. It is noted that the gate cut feature **211** depicted in FIG. **14C-1** is defined by the width $w_{sub.2}$ throughout the entire length along Z direction, similar as the embodiments depicted in FIGS. **5B-1** and **6B-1**.

(50) At operation **122**, still referring to **14A** to **14C**, method **100** performs additional processing steps to the device **200**. For example, the method **100** forms a dielectric layer **272** over the metal gate stack **230**, thereby filling the gate trench. In some embodiments, the dielectric layer **272** is configured to provide etching selectivity during subsequent fabrication processes including, for example, patterning the ILD layer **242** to form S/D contact openings (not shown) over the S/D features **224**. Accordingly, the dielectric layer **272** has a composition different from that of the ILD layer **242**. In some embodiments, the dielectric layer **272** includes SiN, SiCN, SiOC, SiON, SiOCN, other suitable materials, or combinations thereof. The dielectric layer **272** may be deposited by any suitable method, such as ALD, CVD, PVD, other suitable methods, or combinations thereof. Subsequently, the method **100** removes portions of the dielectric layer **272** formed over the ILD layer **242** in one or more CMP process, thereby planarizing the top surface of the device **200**.

(51) The additional processing steps may further include forming an S/D contact **228** over the S/D features **224**. The S/D contact **228** may include any suitable conductive material, such as Co, W, Ru, Cu, Al, Ti, Ni, Au, Pt, Pd, and/or other suitable conductive materials. The method **100** may form an S/D contact opening (or trench) in the ILD layer **242** via a series of patterning and etching processes and subsequently deposit a conductive material in the S/D contact opening using any suitable method, such as CVD, ALD, PVD, plating, and/or other suitable processes. In some

embodiments, a silicide layer (not depicted) is formed between the S/D features **224** and the S/D contact **228**. The silicide layer may include nickel silicide, cobalt silicide, tungsten silicide, tantalum silicide, titanium silicide, platinum silicide, erbium silicide, palladium silicide, other suitable silicide, or combinations thereof. The silicide layer may be formed over the S/D feature **224** by a series of deposition, thermal, and etching processes.

(52) The additional fabrication process **132** to the device **200** may also include forming a multi-layer interconnect (MLI) structure (not depicted) thereover. The MLI may include various interconnect features, such as vias and conductive lines, disposed in dielectric layers, such as ESLs and ILD layers **242**. In some embodiments, the vias are vertical interconnect features configured to interconnect a device-level contact, such as the S/D contact **228** or a gate contact (not depicted), with a conductive line, or interconnect different conductive lines, which are horizontal interconnect features. The ESLs and the ILD layers of the MLI may have substantially the same compositions as those discussed above with respect to the ESL described above and the ILD layer **242**, respectively. The vias and the conductive lines may each include any suitable conductive material, such as Co, W, Ru, Cu, Al, Ti, Ni, Au, Pt, Pd, a metal silicide, other suitable conductive materials, or combinations thereof, and be formed by a series of patterning and deposition processes. Additionally, each via and conductive line may additionally include a barrier layer that comprises TiN and/or TaN.

(53) Although not intended to be limiting, one or more embodiments of the present disclosure provide many benefits to a semiconductor device and the formation thereof. For example, the present disclosure provides a method of forming a gate cut feature with high aspect ratio. The gate cut feature is formed prior to the forming of the dummy gate stacks. The gate cut feature is disposed between two fins and connects to the isolation feature through an opening in the cladding layer of the fins. Embodiments of the disclosed methods can be readily integrated into existing processes and technologies for manufacturing NS FETs, FinFETs, and/or other suitable devices.

(54) In one aspect, the present disclosure provides a method. The method includes forming a fin protruding from a substrate, forming a first dielectric feature adjacent to the fin over the substrate, forming a cladding layer over the fin and the first dielectric feature, removing a portion of the cladding layer to form an opening, the opening exposing the first dielectric feature, and forming a second dielectric feature adjacent to the cladding layer, the second dielectric feature filling the opening, forming a dummy gate stack over the fin and the second dielectric feature, forming source/drain (S/D) features in the fin adjacent to the dummy gate stack, and replacing the dummy gate stack and the cladding layer with a metal gate stack, the second dielectric feature dividing the metal gate stack. In some embodiments, the removing of the portion of the cladding layer forms the opening defined by a width that is less than a width of the second dielectric feature. In some embodiments, the removing of the portion of the cladding layer forms the opening defined by a width equals to a width of the second dielectric feature. In some embodiments, the metal gate stack has a top surface coplanar with a top surface of the second dielectric feature. In some embodiments, the method further includes forming a dielectric helmet (**214**) over the second dielectric feature, the dielectric helmet separating a top portion of the metal gate stack. In some embodiments, the method further includes forming a third dielectric feature disposed on an opposite side of the fin away from the second dielectric feature, the third dielectric feature being disposed over the cladding layer without an opening to expose the first dielectric feature, and the cladding layer under the third dielectric feature is replaced by the metal gate stack without being separated by the third dielectric feature. In some embodiments, a portion of the metal gate stack is disposed over the third dielectric feature.

(55) In another aspect, the present disclosure provides a method of forming a semiconductor structure. The method includes forming a first fin and a second fin disposed over a substrate, each of the first and the second fins including alternating channel layers and non-channel layers, forming an isolation feature between the first fin and the second fin over the substrate, forming a cladding

layer over the first fin, the second fin, and the isolation feature, forming an opening in the cladding layer over the isolation feature, thereby separating the cladding layer into a first portion over the first fin and a second portion over the second fin, forming a gate cut feature between the first portion and the second portion of the cladding layer, thereby filling the opening, forming a dummy gate stack over the first fin, the second fin, and the gate cut feature, forming source/drain (S/D) features in the first and second fin adjacent to the dummy gate stack, and replacing the dummy gate stack, the cladding layer, and the non-channel layers with a metal gate stack. The gate cut feature truncates the metal gate stack into two portions disposed on opposite sides of the gate cut feature. In some embodiments, the forming of the opening in the cladding layer removes an entire horizontal portion of the cladding layer between the first and the second fin. In some embodiments, the forming of the opening in the cladding layer removes a portion of a horizontal portion of the cladding layer between the first and the second fin. In some embodiments, the isolation feature includes a material that is the same as a material included in the gate cut feature. In some embodiments, the metal gate stack has a top surface coplanar with a top surface of the gate cut feature. In some embodiments, the method further includes forming a dielectric helmet over the gate cut feature, the dielectric helmet cutting the metal gate stack into two portions disposed on opposite sides of the dielectric helmet. In some embodiments, the method further includes forming a third fin disposed over the substrate, the first fin and the third fin being disposed on opposite sides of the second fin, and forming a dummy gate cut feature between the second and the third fin, a third portion of the cladding layer over the third fin being connect to the second portion of the cladding layer, and the metal gate stack under the dummy gate cut feature being continuous. In some embodiments, a portion of the metal gate stack spans across the second fin, the dummy gate cut feature, and the third fin.

(56) In yet another aspect, the present disclosure provides a semiconductor structure. The semiconductor structure includes a first and a second semiconductor fin each including a stack of semiconductor layers and disposed over a substrate, an isolation feature between the first and the second semiconductor fin and disposed over the substrate, a gate stack interposed with the first and the second semiconductor fin and disposed over the isolation feature, and a gate cut feature separating the gate stack into a first portion over the first semiconductor fin and a second portion over the second semiconductor fin, wherein a bottom surface of the gate cut feature contacts the isolation feature through a hole in the gate stack. In some embodiments, the semiconductor structure further includes a third semiconductor fin disposed next to the second semiconductor fin in a direction away from the first semiconductor fin, and a dummy gate cut feature disposed between the second semiconductor fin and the third semiconductor fin, the dummy gate cut feature being separated from the isolation feature by the gate stack. In some embodiments, a first portion of the gate stack over the first semiconductor fin is separated from a second portion of the gate stack over the second semiconductor fin; and a third portion of the gate stack over the third semiconductor fin contacts the second portion. In some embodiments, a top surface of the gate stack is coplanar with a top surface of the gate cut feature. In some embodiments, the semiconductor structure further includes a dielectric helmet over the gate cut feature, wherein the dielectric helmet is disposed between the first portion and the second portion of the gate stack.

(57) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method, comprising: forming a fin protruding from a substrate; forming a first dielectric feature adjacent to the fin over the substrate; forming a cladding layer over the fin and the first dielectric feature; removing a portion of the cladding layer to form an opening, wherein the opening exposes the first dielectric feature; and forming a second dielectric feature adjacent to the cladding layer, wherein the second dielectric feature fills the opening; forming a dummy gate stack over the fin and the second dielectric feature; forming source/drain (S/D) features in S/D regions of the fin not covered by the dummy gate stack; removing the dummy gate stack to form a gate trench, the gate trench exposing the cladding layer; removing the cladding layer; and forming a metal gate stack in the gate trench, wherein the second dielectric feature divides the metal gate stack.
2. The method of claim 1, wherein the removing of the portion of the cladding layer forms the opening defined by a width that is less than a width of the second dielectric feature.
3. The method of claim 1, wherein the removing of the portion of the cladding layer forms the opening defined by a width equals to a width of the second dielectric feature.
4. The method of claim 1, wherein the metal gate stack has a top surface coplanar with a top surface of the second dielectric feature.
5. The method of claim 1, further comprising: forming a dielectric helmet (214) over the second dielectric feature, wherein the dielectric helmet separates a top portion of the metal gate stack.
6. The method of claim 1, further comprising: forming a third dielectric feature disposed on an opposite side of the fin away from the second dielectric feature, wherein the third dielectric feature is disposed over the cladding layer without an opening to expose the first dielectric feature, and wherein the cladding layer under the third dielectric feature is replaced by the metal gate stack without being separated by the third dielectric feature.
7. The method of claim 6, wherein a portion of the metal gate stack is disposed over the third dielectric feature.
8. A method of forming a semiconductor structure, comprising: forming a first fin and a second fin disposed over a substrate, wherein each of the first and the second fins includes alternating channel layers and non-channel layers; forming an isolation feature between the first fin and the second fin over the substrate; forming a cladding layer over the first fin, the second fin, and the isolation feature; forming an opening in the cladding layer over the isolation feature, thereby separating the cladding layer into a first portion over the first fin and a second portion over the second fin; forming a gate cut feature between the first portion and the second portion of the cladding layer, thereby filling the opening; forming a dummy gate stack over the first fin, the second fin, and the gate cut feature; forming source/drain (S/D) features in regions of the first and second fins not covered the dummy gate stack; removing the dummy gate stack to form a gate trench, the gate trench exposing the cladding layer; removing the cladding layer to expose sidewalls of the non-channel layers in the gate trench; removing the non-channel layers; and forming a metal gate stack in the gate trench, wherein the gate cut feature truncates the metal gate stack into two portions disposed on opposite sides of the gate cut feature.
9. The method of claim 8, wherein the forming of the opening in the cladding layer removes an entire horizontal portion of the cladding layer between the first and the second fin.
10. The method of claim 8, wherein the forming of the opening in the cladding layer removes a portion of a horizontal portion of the cladding layer between the first and the second fin.
11. The method of claim 8, wherein the isolation feature includes a material that is the same as a material included in the gate cut feature.
12. The method of claim 8, wherein the metal gate stack has a top surface coplanar with a top surface of the gate cut feature.

13. The method of claim 8, further comprising: forming a dielectric helmet over the gate cut feature, wherein the dielectric helmet cuts the metal gate stack into two portions disposed on opposite sides of the dielectric helmet.

14. The method of claim 8, further comprising: forming a third fin disposed over the substrate, wherein the first fin and the third fin are disposed on opposite sides of the second fin; and forming a dummy gate cut feature between the second and the third fin, wherein a third portion of the cladding layer over the third fin is connect to the second portion of the cladding layer, and wherein the metal gate stack under the dummy gate cut feature is continuous.

15. The method of claim 14, wherein a portion of the metal gate stack spans across the second fin, the dummy gate cut feature, and the third fin.

16. A method, comprising: forming a fin over a substrate, the fin including alternating channel layers and sacrificial layers; depositing an isolation feature on sidewalls of the fin; forming a cladding layer over the sidewalls of the fin and a top surface of the isolation feature; forming an isolation structure over a horizontal portion of the cladding layer, wherein the fin and the isolation structure sandwich a vertical portion of the cladding layer, wherein a top surface of the vertical portion of the cladding layer is not covered by the isolation structure; removing the vertical portion and the horizontal portion of the cladding layer to form a trench, the trench exposing sidewalls of the sacrificial layers, a bottom portion of the trench extending to a position directly under the isolation structure; removing the sacrificial layers; and forming a metal gate structure in the trench, the metal gate structure wrapping around each of the channel layers, wherein a portion of the metal gate structure filling the bottom portion of the trench is directly under the isolation structure.

17. The method of claim 16, wherein, in a cross section perpendicular to a lengthwise direction of the channel layers, the metal gate structure fully surrounds the isolation structure.

18. The method of claim 16, further comprising: forming an opening in the horizontal portion of the cladding layer, wherein the isolation structure extends downwardly through the opening and has a physical contact with the top surface of the isolation feature.

19. The method of claim 18, further comprising: recessing the metal gate structure, such that a top surface of the metal gate structure is lower than a top surface of the isolation structure.

20. The method of claim 16, further comprising: forming a dummy gate stack over the fin and the isolation structure; patterning the dummy gate stack to form an opening directly above the isolation structure; recessing the isolation structure through the opening; and removing the dummy gate stack.
